

# KUO-HAN HUNG

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🏠 [khhung-906.github.io](https://github.com/khhung-906)

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## Education

### Stanford University

June 2027 (Expected)

*M.S. in Computer Science*

*Stanford, CA*

- GPA: 4.0/4.0; Specialization: Artificial Intelligence and Robotics.

### National Taiwan University

Dec. 2024

*B.S. in Computer Science and Information Engineering*

*Taipei, Taiwan*

- GPA: 4.24/4.3, Top 3% of Class. **Awards:** Bachelor Thesis Dean's Award, Presidential Award (×2), Dean's List (×3).

## Technical Skills

- **Robotics & Systems:** Real-Robot Deployment and Training, Real-to-Sim, Real-Time Control, ROS, Docker, Linux.
- **Parallel & Languages:** Distributed Training, Ray, SLURM, GCP, Python, C/C++, Bash, PyTorch, JAX, NumPy.

## Research Experience

### Stanford Artificial Intelligence Laboratory (ILIAD & IRIS Labs)

Sept. 2025 – Present

*Graduate Researcher, Advisors: Prof. Dorsa Sadigh & Prof. Chelsea Finn*

*Stanford, CA*

- **RL finetuning of real-time VLA policies.** Decoupled slow VLA proposals from a fast edit policy, raising success from 42% to 97% across six tasks including dexterous light-string routing, with under 10 min of online data. [1]
- **Sample-efficient RL finetuning of pretrained VLAs.** Built a stable system and algorithm to improve VLA with residual action edits and Q-guided sampling, reaching 30/30 success with 19 min of online data (100+ GitHub stars). [2]
- **Scaling Q-functions** with Transformers in offline RL. Introduced attention entropy regularization to stabilize and scale Transformer value functions, gaining 43% as model size grows where all prior methods fail. [3]
- **Visual representation pretraining** for reinforcement learning (ongoing).

### Robot Learning Lab

Oct. 2024 – May 2025

*Undergraduate Researcher, Advisor: Prof. Shao-Hua Sun*

*Taipei, Taiwan*

- **Manipulation policies learned from action-free videos**, using optical flow as an action surrogate. [5]

### Communication and Multimedia Lab

May 2022 – May 2024

*Undergraduate Researcher, Advisors: Prof. Winston H. Hsu & Prof. Yi-Ting Chen*

*Taipei, Taiwan*

- **Vision-instruction correlation rewards** for long-horizon manipulation: 43% over prior best. [4]
- **Behavior error detection** in few-shot imitation policies via representation patterns.

## Work Experience

### IBM Research

June 2024 – Sept. 2024

*Research Intern, Host: Dr. Pin-Yu Chen*

*Yorktown Heights, NY*

- Found that attention drifts from the original instruction during **prompt injection attacks on LLM agents**.
- Traced it to specific attention heads and built a detector with a 31% AUROC gain over prior best.

### Microsoft

July 2023 – June 2024

*Applied Scientist Intern, Host: Dr. Chiqun Zhang*

*Mountain View, CA (Remote)*

- **Integrated Bing Maps with Microsoft Copilot** for conversational, multi-destination map queries.
- Formulated a pathfinding algorithm that learns from history, **achieving a 10× speedup over Dijkstra**.

## Selected Publications

(\* indicates equal contribution)

- [1] **Reinforcement Learning for Real-Time Vision-Language-Action Policies**  
Perry Dong\*, **Kuo-Han Hung\***, Dorsa Sadigh, and Chelsea Finn. Under Review
- [2] **EXPO-FT: Sample-Efficient Reinforcement Learning Finetuning for Vision-Language-Action Models**  
Perry Dong\*, **Kuo-Han Hung\***, Tian Gao, Dorsa Sadigh, and Chelsea Finn. CoRL 2026
- [3] **TQL: Scaling Q-Functions with Transformers by Preventing Attention Collapse**  
Perry Dong\*, **Kuo-Han Hung\***, Alexander Swerdlow, Dorsa Sadigh, and Chelsea Finn. ICML 2026
- [4] **VICtoR: Learning Hierarchical Vision-Instruction Rewards for Long-horizon Manipulation**  
**Kuo-Han Hung\***, Pang-Chi Lo\*, Jia-Fong Yeh\*, Han-Yuan Hsu, Yi-Ting Chen, and Winston Hsu. ICLR 2025
- [5] **Learning Skill Abstraction from Action-Free Videos via Optical Flow**  
Hung-Chieh Fang\*, **Kuo-Han Hung\***, ..., Min-Hung Chen, and Shao-Hua Sun. ICML Workshop 2025